

Algorithmic Lending and Model Risk

A Fairness, Transparency, and Data-Privacy Audit of Machine-Learning Credit Scoring

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Abstract

Artificial Intelligence (AI) has grown rapidly over the last few years, spanning almost every sector. The banking sector has embraced AI and incorporated it into daily banking activities. While AI has immense potential to improve the customer banking experience, its inherent issues are often overlooked. This study develops a credit-scoring model using Machine Learning (ML) algorithms to determine whether to grant a loan to a customer. The models are trained on public data and later evaluated for fairness, transparency, and data privacy in ML lending decisions. They are also mapped with the RBI's Digital Lending Directions (2025), the RBI FREE-AI Committee Framework, and the Digital Personal Data Protection (DPDP) Act (2023).

1. Introduction

Lending has increasingly become automated. RBI's Digital Lending Directions define digital lending as an automated lending process involving customer assessment, credit assessment, loan approval, and associated customer service. When an ML model decides who will be approved, three potential issues need to be checked: first, whether the model treats comparable applicants from different groups unequally; second, whether the model can justify its decision; and third, whether it uses large volumes of personal data whose collection and retention could be legally constrained. This concern stems from the black-box nature of major AIs, where it is hard to determine which factors drive their decisions. ML models require human data to train, and the sharing of personal data raises concerns about breaches of data privacy and unfair business conduct, which RBI highlights in its Lending Directions as a core concern motivating the regulations.

This paper does not propose that AI should not be used for lending. Rather, it argues for using AI without complacency, with close attention to fairness, transparency, and data privacy. It aims to highlight the limitations of AI so that decisions are made with much greater calibration.

2. Methodology

The model was trained on the Statlog German Credit dataset, which contains data of 1000 past borrowers, each labeled as a "good" or "bad" credit risk. Each borrower was described by 20 attributes, including account status, loan amount and duration, employment, housing, age, and so on. Out of 1000 borrowers, 700 were classified as "good" and 300 as "bad".

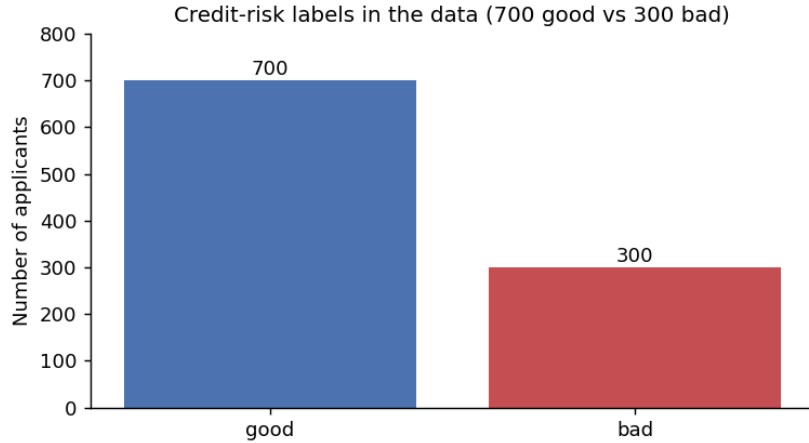


Figure 1: Credit-risk labels in the data (700 good vs. 300 bad).

This information is crucial in testing the accuracy of our model. It is so because a model that only says “good” would be correct in 70% of cases anyway. To preserve this balance, the data were split into training and test sets at a 3:1 ratio; the training set contained 750 applicants, and the test set contained 250. In addition, the same ratio of “good” to “bad” categories was maintained across training and test data by ensuring that 70% of the data in both was “good.”

Some features contain numbers and others are text. Because the numerical ranges varied across features, I used standard scaling to bring all numerical values to a common scale. For features containing text, I used one-hot encoding, which converts text into a numerical format the model can understand. After one-hot encoding, the overall features reached 61.

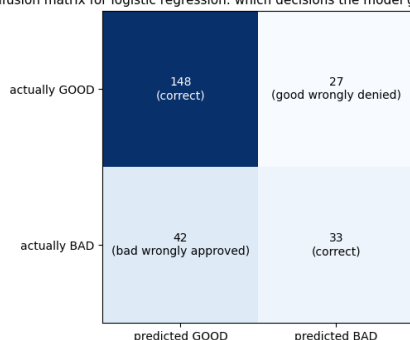
The same data was trained on two different models: Logistic Regression and Random Forest. Logistic Regression assigns weights to each attribute and assigns a score to determine the category to which an applicant belongs. Random Forest, on the other hand, combines hundreds of branching decision rules to decide whether an applicant should lie in the “good” category or the “bad” category. These two inherently different models were chosen to show that the issues of fairness, opacity, and data privacy were not model-dependent.

3. Findings

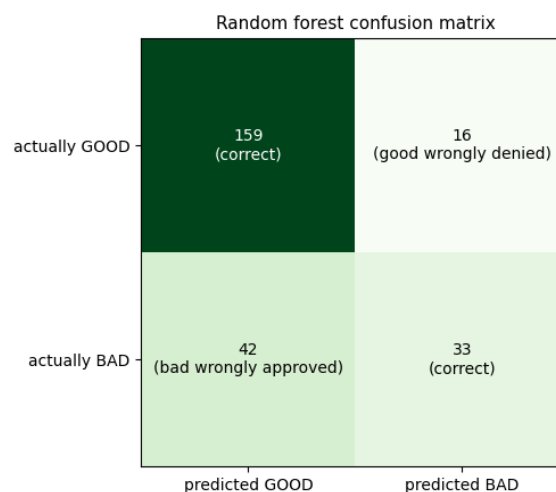
3.1 Accuracy vs. Real-World Efficacy

Testing the trained model on test data shows that Logistic Regression achieved 72.4% accuracy, whereas Random Forest achieved 76.8% accuracy. The higher accuracy mainly came from approving more good applicants: Logistic Regression denied 27 of 175 good applicants, and Random Forest denied 16 of 175. Both models approved 42 of 75 bad loans, so the false-positive rate was equal. This shows that higher accuracy did not improve the effectiveness of bad-loan screening. Both models missed more than half of all bad borrowers, underscoring the need for human review at the approval stage.

Confusion matrix for logistic regression: which decisions the model gets right and wrong



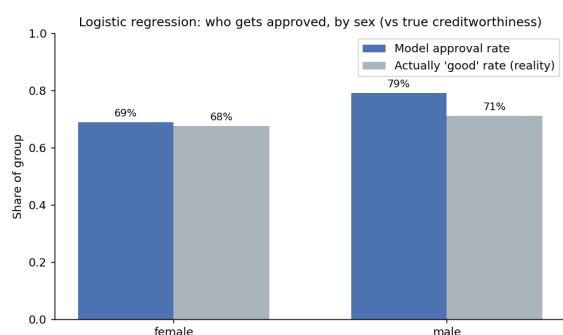
(a) Confusion matrix for Logistic Regression.



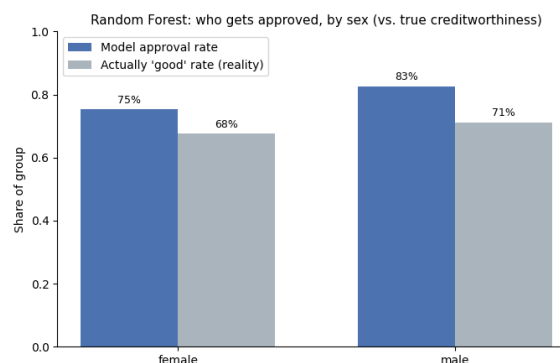
(b) Confusion matrix for Random Forest.

3.2 Fairness and the Amplification of Bias

When I tested how various decisions are driven by both models, the following risks were highlighted. Splitting the test applicants by sex showed that 68% of female test applicants were in the “good” category, and 71% of male test applicants were in the “good” category. The Logistic Regression model classified 69% of female test applicants as “good” and 79% of male test applicants as “good”. This widened the approval gap between females and males to 10%, from 3% originally. Random Forest classified 75% of females and 83% of males as “good”. These results show that the models widen the advantage for men over women.



(a) Logistic Regression: approval by sex vs. true creditworthiness.



(b) Random Forest: approval by sex vs. true creditworthiness.

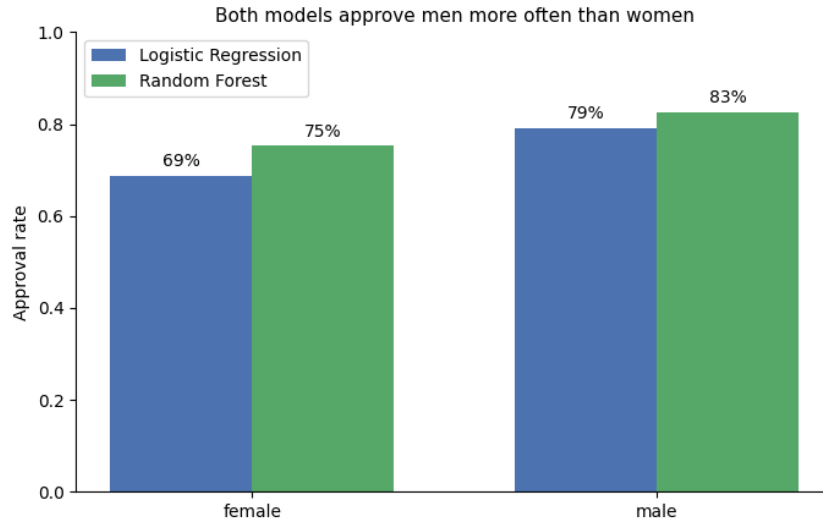
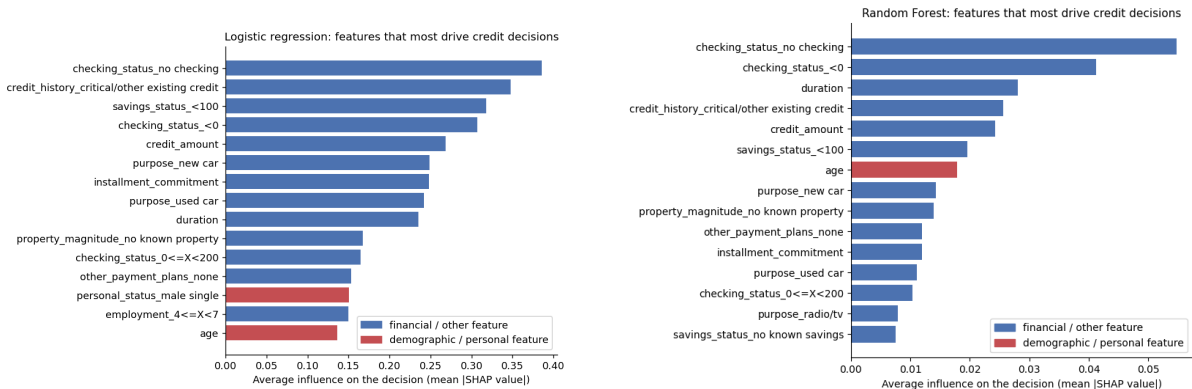


Figure 4: Both models approve men more often than women.

A better metric to look at here is the true positive rate (TPR), which shows, among the people who are genuinely good credit risks, the share that the model correctly approves. TPR was 77% for females and 88% for males, as determined by Logistic Regression. For Random Forest, it was 87% for females and 93% for males. This means that, among applicants who are actually good, a higher percentage of males were approved than females. This shows unfairness because when a person comes under the “good” category, i.e., they deserve to get the loan, they will more likely get the loan approval if they are male than if they are female.

3.3 Opacity and the “Black Box” Problem

ML models act like a black box, so we rarely know how they work internally. As a result, a lender may be unable to explain how the ML model made its decision to accept or reject a loan application. To overcome this issue, SHAP (Shapley Additive exPlanations) was used, which assigns values to each feature. It shows the impact each attribute has on the overall result and ranks them, with the top-ranked feature having the greatest impact. SHAP helps, to a certain extent, overcome this issue by ranking different features. SHAP found that 2 of the top 15-ranked features in Logistic Regression were demographic. Even in Random Forest, where only 1 of the top 15 features was a demographic feature, it ranked 7, above the feature “installment commitment,” which contains information on loan repayment as a share of income.



(a) Logistic Regression: top SHAP features.

(b) Random Forest: top SHAP features.

To investigate what was driving the difference in male and female results, SHAP can be used.

Running SHAP on Logistic Regression showed that, out of 61 features (54 financial factors), “personal_status_male single” was ranked 13, and “personal_status_female div/dep/mar” was ranked 28. Even for Random Forest, “personal_status_female div/dep/mar” was ranked 17, and “personal_status_male single” was ranked 19. This shows that the model gives greater weight to gender even when other financial data are present.

3.4 Proxy Trap and Data Privacy

An intuitive solution to the above problem is to remove the feature that contains gender information. However, when the model was trained without gender information and other features were used to predict the applicant’s gender, it achieved an AUC score of 0.65, which is better than a coin flip’s 0.5. This shows that neutral features such as “housing”, “number of dependents,” and “installment commitment,” among others, still carry information about sex. The chart below shows the ranking of how much various features act as proxies for gender data. Thus, removing the feature containing gender information provided only false relief.

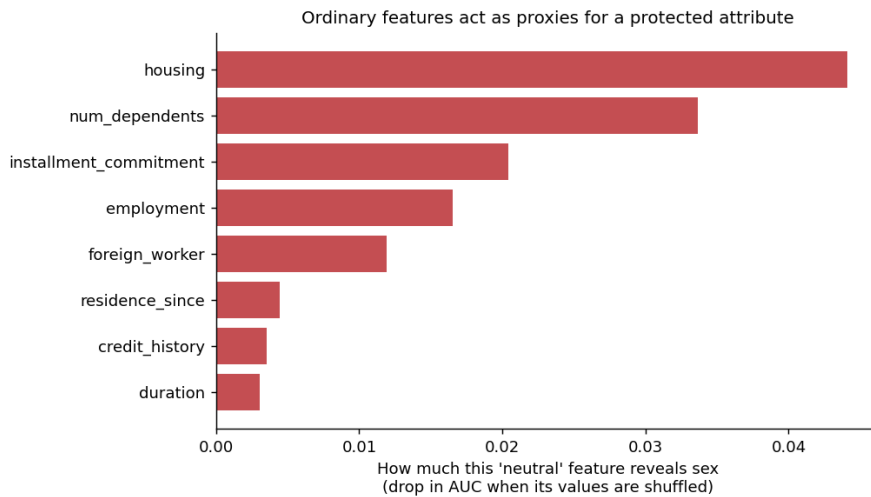


Figure 6: Ordinary features act as proxies for a protected attribute.

The model was trained on demographic features including age, gender, marital status (single, married, or divorced), and whether a person is a foreign worker. These are personal, sensitive information of an applicant, which is vulnerable to data breaches. As seen before, even if these features are removed, they reappear through their proxies. This highlights the privacy concern associated with training ML models on personal data. The DPDP Act, 2023, restricts the use of personal data to lawful, specific purposes and requires organizations to obtain clear, unambiguous consent before processing information.

4. Regulatory Mapping

I have mapped the three major issues of fairness, opacity, and privacy with the following frameworks to show how the findings support the paper’s central argument.

4.1 RBI Digital Lending Direction, 2025

Creditworthiness Assessment: Lenders must objectively evaluate creditworthiness based on a borrower’s verified economic profile—specifically age, occupation, and income. The exact basis for this evaluation must be documented for regulatory audits. Furthermore, lenders are strictly prohibited from automatically increasing credit limits without the borrower’s explicit, recorded consent.

Consistent Treatment: Aggregator platforms partnering with multiple lenders must use documented matching algorithms that apply uniformly to similar borrowers. Lending Service Providers are forbidden from using deceptive “dark patterns” to unfairly steer customers toward specific lenders. All eligible loan offers must be presented objectively and without bias.

Data Minimization: Lending apps must practice strict data minimization, collecting only essential information with explicit, granular consent. Accessing personal mobile resources, such as contact lists or media files, is banned. Borrowers retain full control, possessing the right to deny data requests, revoke consent, and demand complete data deletion.

4.2 The RBI FREE-AI Committee: Seven Sutras

The RBI FREE-AI committee provides seven sutras, each with a guiding principle.

Sutra 4 — Fairness and Equity: Provides that AI outcomes should be non-discriminatory. It calls for a fairness test.

Sutra 5 — Accountability: Rests on the entity deploying the AI, regardless of the level of automation. The lender cannot outsource responsibility to a vendor’s model.

Sutra 2 — People First: States that AI should aid decision-making but defer to human judgment. As both ML models miss over half of true defaulters, they should serve as decision aids under human oversight, not as independent entities.

4.3 Digital Personal Data Protection Act, 2023

Purpose limitation and consent (Sections 5–6): Under Sections 5 and 6, personal data processing requires consent that is free, specific, informed, unconditional, and unambiguous, with clear affirmative action. It must be limited to the specified purpose, minimizing privacy risks associated with collecting unnecessary attributes.

Completeness, accuracy, and consistency (Section 8(3)): Under Section 8(3), when personal data is used to make a decision affecting an individual, the Data Fiduciary must ensure its accuracy, completeness, and consistency. This statutory obligation ensures uniform, reliable data quality across all credit-decisioning models.

Correction and erasure (Sections 12 and 8(7)): Section 12 grants individuals the right to demand data correction and erasure. However, lending data pipelines must account for Section 8(7), which dictates that erasure rights do not apply if retaining the data is necessary to comply with other prevailing laws.

5. Recommendations

To ensure AI in lending aligns with India’s regulatory frameworks, banks must move beyond mere accuracy metrics and operationalize the following safeguards:

- **Mandate Demographic Stress-Testing:** A model’s performance must not be judged solely on overall accuracy. Prior to and during deployment, banks must test how ML models perform across specific demographic groups to ensure approval rates accurately reflect actual creditworthiness.
- **Audit for Proxy Variables:** Deleting protected categories (like gender) from a dataset provides only false relief. Lenders must actively and continuously audit the remaining “neutral” data to ensure that everyday variables are not serving as proxies for protected traits.
- **Enforce Explainability Trade-offs:** The choice of ML models must heavily weigh explainability. If a bank deploys an opaque “black box” model (e.g., Random Forest) over a linear one because of a slight accuracy gain, it must explicitly document why the trade-off is justified and implement a robust system (e.g., SHAP) to explain adverse decisions to borrowers.

- **Implement “Human-in-the-Loop” Oversight:** We must never let the ML model have the final say. AI must be positioned strictly as an advisory tool. The final decision-maker must retain the authority to review, explain, and override the model’s acceptance or rejection.
- **Strict Data Minimization:** During the onboarding process, data collection algorithms should be restricted to scraping only the fields strictly necessary to assess financial credit-worthiness (e.g., verified income, debt-to-income ratio, or repayment history).

6. Conclusion

With existing frameworks like the FREE-AI Sutras, the RBI’s Digital Lending Directions, and the DPDP Act already in place, the policy focus should not be on restricting the use of AI in banking. Rather, the focus must shift to strictly regulating lenders and ensuring the operationalization of the fairness testing, transparent documentation, and human oversight that our current laws already demand. AI has the potential to expand financial inclusion, but only if it is deployed without complacency.